EXERCISE 4: SQL JOINS

Question 1: Inner Join

: Join Students and Grades to display Only Student who have grades.

```
SELECT S.Student_id, Student_name,
        Grade
From Students As S
Inner join Grade As G
On S.student_id = G.Student_id;
```

Student_id	Student Name	Grade
2	Bob	B
3	Charlie	A

3

Question 2: LEFT JOIN

: Display all employees and the department they belong to. Include employees with no depastment.

```
SELECTEEmp_id, Emp_Name,
        Dept_name
From Employees AS E
LEFT Join Department AS D
On E.Emp_id = D.Emp_id;
```

Emp_id	Emp_Name	Dept_Name
1	John	Null
2	Lisa	HR
3	Mike	Null

Question 3 : Full Outer Join

: Display a complete list of products & their quantities Sold. Include products with ~~their~~ no sales & Sales for Unknown product

```
SELECT P.Product_id, Product_name,
       Quantity
FROM Products AS P
Full OUTER JOIN Sales AS S
On P.Product_id = S.Product_id;
```

Product id	Product Name	Quantity
1	Laptop	Null
2	Mouse	50
3	Keyboard	Null
4	Null	30

Question 4 : LEFT JOIN & CASE

: Display all orders & ~~indicate~~ indicate whether the Customer is 'New or Returning'

```
SELECT Order_id, Customer_id,
       Amount, Customer_Name,
      Count(Order_id) AS Order_Count,
      CASE
          WHEN Order_Count =< 1 THEN 'New'
          ELSE 'Returning'
      END AS Customer_Type
FROM Orders AS O
LEFT JOIN Customers AS C
On O.Order_id = C.Order_id
GROUP BY Order_id, Customer_id, Customer_Type,
         Amount, Customer_name;
```

Order id	Customer id	Amount	Customer Name	Customer type	Order
1	101	500	Paul	New	1
2	102	300	Sarah	New	1
3	105	0	Null	New	1

Question 5 : LEFT Join + Group By + Sum

: Show total sales per region & Include regions with no sales

```
SELECT Region_id, Region_name,
        Sum (Amount) AS Total_Sales
FROM Sales AS S
FULL OUTER Join Regions AS R
On S.Region_id = R.Region_id
GROUP BY Region_id, Region_name;
```

Region_id	Region_Name	Total Sales
1	North	2000
2	South	3500
3	East	Null
4	Null	Null

Question 6 : LEFT Join + CASE

: Classify students based on attendance

```
SELECT Student_id, Name,
      Days_present,
    CASE
        WHEN Days_present Between 0
        AND 5 THEN 'Poor_Attendance'
        WHEN Days_present Between 6
      AND 15 THEN 'Needs_Improvement'
        WHEN Days_present > 16 THEN 'Excellent'
```

```
END AS Attendance_Status
FROM Students AS S
LEFT JOIN Attendance AS A
ON S.Student_id = A.Students_id;
```

Student_id	Name	Days Present	Attendance Status
1	Alice	18	Excellent
2	Bob	5	Poor Attendance
3	Charlie	Null	Poor Attendance

Question 7: INNER JOIN + COUNT + GROUP BY

: Show number of tasks per project. Only include projects that have tasks.

```
SELECT Project_id, NAME,
       Count (task_id) AS Task_Count
FROM Projects AS P
INNER JOIN Tasks AS T
ON P.Project_id = T.Product_id
GROUP BY Project_id, Name;
```

Project_id	Name	Task Count
1	AI Chatbot	2
2	Website	1
3	Data Report	Null

Question 8 : Full Outer Join + Case + Where

: Classify Customers based on whether they returned anything and filter by high Order total

```
SELECT Cust_id, Order_Total,
       Return_total,
    CASE
        WHEN Return_Total is null THEN 'No Return'
        WHEN Return_Total = 0 THEN 'No_Return'
        ELSE 'Returned'
        END AS Return_Status
FROM Orders AS O
Full Outer Join Returns AS R
On O.Cust_id = R.Cust_id
WHERE Order Total > 150;
```

Cust ID	Order total	Return total	Return Status
12	250	Null	No_Return
13	180	Null	No_Return

Question 9 : Left Join + Count + ORDER BY

: Count how many times each user logged in

```
SELECT User_id, Name,
       Count( Login date ) AS login Count
FROM Users AS U
LEFT Join Logins AS L
On U.User_id = L.User_id
GROUP BY User_id, Name
ORDER BY Login Count DESC;
```

User_id	Name	Login Count
2	Gloria	2
3	Steve	1
1	Nelson	Null

Question 10 : LEFT JOIN + CASE + ORDER BY
: Show all teachers & the subjects they teach. If no subject, label appropriately

```
SELECT Teacher_id, teacher name,
        ~~Subject_name~~,
    CASE
        WHEN Subject Name Is Null
        THEN 'Appropriately'
        ELSE 'Subject_name'
    END AS 'Subject_name'
FROM Teacher As T
LEFT Join Subjects As S
~~On T.Teacher_id  S.Teaching~~
On T.Teacher_id = S.Teacher_id
ORDER BY Teacher_name Asc;
```

Teacher id	Teacher Name	Subject Name
~~Ms~~ 2	Ms. Ndaba	Appropriately
3	Mr. Dlamini	Appropriately
1	Mr. Hlongwane	Math
1	Mr. Hongwane	Science